Zhang, Y., Diddee, H., Holm, S., Liu, H., Liu, X., Samuel, V., Wang, B., and Ippolito, D. Noveltybench: Evaluating language models for humanlike diversity. 2025. URL https://api.semanticscholar.org/CorpusID:277621515.

Zhao, Y., Zhang, R., Li, W., Huang, D., Guo, J., Peng, S., Hao, Y., Wen, Y., Hu, X., Du, Z., Guo, Q., Li, L., and Chen, Y. Assessing and understanding creativity in large language models. *ArXiv*, abs/2401.12491, 2024. URL https://api.semanticscholar.org/CorpusID:267094860.

Zhong, S., Huang, Z., Gao, S., Wen, W., Lin, L., Zitnik, M., and Zhou, P. Let's think outside the box: Exploring leap-of-thought in large language models with creative humor generation. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition, CVPR 2024*, 2024.

Zhu, Y., Lu, S., Zheng, L., Guo, J., Zhang, W., Wang, J., and Yu, Y. Texygen: A benchmarking platform for text generation models. *The 41st International ACM SIGIR Conference on Research & Development in Information Retrieval*, 2018.

A. Transformer Training Objectives

Let $\texttt{LM}_\theta$ be our language model, parameterized by θ, for which $\texttt{LM}_\theta(\hat{s}_i = s_i; \boldsymbol{s}_{<i})$ is the probability it assigns to the ith output $\hat{s}_i$ being s_i, given as input a sequence $\boldsymbol{s}_{<i}$. Let $(\boldsymbol{p}, \boldsymbol{r})$ be a prefix-response pair. In standard next-token finetuning, we maximize the objective:

$$\mathcal{J}_{\texttt{next-token}}(\theta) = \mathbb{E}_{\mathcal{D}}\Big[\sum_{i=1}^{L_{\text{resp}}} \log \texttt{LM}_\theta\left(\hat{r}_i = r_i; \boldsymbol{p}, \boldsymbol{r}_{<i}\right)\Big] \tag{2}$$

In teacherless (multi-token) training (Monea et al., 2023; Bachmann & Nagarajan, 2024; Tschannen et al., 2023), we make use of an uninformative input string $\$$ that simply corresponds to a series of dummy tokens $\$$.

$$\mathcal{J}_{\texttt{multi-token}}(\theta) = \mathbb{E}_{\mathcal{D}}\Big[\sum_{i=1}^{L_{\text{resp}}} \log \texttt{LM}_\theta\left(\hat{r}_i = r_i; \boldsymbol{p}, \$_{<i}\right)\Big] \tag{3}$$

B. Further discussion

B.1. Style of noise-injection

Our technique of injecting noise into the model is somewhat different from how noise is introduced in traditional VAEs (Kingma & Welling, 2014) or GANs (Goodfellow et al., 2020), and this difference is worth noticing. In traditional approaches, although the model learns a noise-output mapping, this mapping is enforced only at a distribution level i.e., the distribution of noise vectors must map to a distribution of real vectors. However, in our approach we arbitrarily enforce what noise vector goes to what real datapoint, at a pointwise level. This raises the open questions of why seed-conditioning works in the first place — surprisingly, without breaking optimization or generalization — and whether there is a way to enforce it at distribution-level, and whether that can provide even greater improvements.

B.2. In-weights vs in-context graphs for combinational creativity

Combinational creativity requires searching through known entities. In abstracting this, there is an interesting choice to be made as to whether the relevant search space is retrieved and spelled out in-context or whether it remains in-weights (like in `Sibling Discovery` and `Triangle Discovery`). We argue that the in-context version does not capture the creative skills required in many real-world tasks. For instance, discovering a fresh and surprising analogy necessitates noticing similarities from sufficiently distinct parts of one's vast, rich space of memory. Thus, the core challenge here lies in retrieving from the entirety of one's memory. If one were to faithfully simulate this an in-context version of this in a model, one would have to provide the entirety of the model's pretraining data in context.

B.3. Examples of `Triangle Discovery`

Although we presented this task as a more complex, higher-order counterpart to `Sibling Discovery`, we retrospectively identify some real-world examples that resemble the higher-order search skill involved in this task.

1. **Discovering contradictions:** Consider identifying non-trivial contradictions within (a large body) of knowledge (like a legal system, or a proof based on many lemmas, or the literature spanning many papers in a certain field). This may require identifying two or more facts that together result in an implication that contradicts another fact.

2. **Discovering feedback loops:** Fields like biology, ecology, climate science and economics may involve discovering non-trivial feedback loops. Unlike feedback loops where two events encourage each other, a non-trivial loop would be one where an `Event A` encourages `Event B`, that in turn encourages `Event C` that in turn encourages `Event A`.

3. **Antanaclasis:** An antanaclasis involves using a word in two different senses in a sentence, while still ensuring that each sense has a coherent relationship with the rest of the sentence. Consider Benjamin Franklin's quote, `Your argument is sound, nothing but sound.` Here, the two senses are $\texttt{sound}_1$ as in "logically correct", $\texttt{sound}_2$ as in "noise". This sentence encodes an pairwise relationship between three entities $\{\texttt{argument}, \texttt{sound}_1, \texttt{sound}_2\}$

individually. While the last two entities (the two senses) themselves must be related to each other (through the common word, `sound`), for a coherent sentence, both senses must also be appropriate descriptors for the first entity, `argument`. Thus, constructing this sentence requires searching through one's vocabulary to discover three words that satisfy these three relationships simultaneously.

4. **Word games:** Some word games require identifying a set of words that simultaneously have pairwise relationships with each other.

   (a) For example, standard crosswords would require identifying sets of 4 or more words that have various simultaneous pairwise intersections in the letters used.

   (b) Devising "& Lit." clues in cryptic crosswords are an altogether different, yet compelling example that require discovering a satisfying triangular relationship. Consider the clue "`Some assassin in Japan`" whose answer is `Ninja`. Here the phrase `Some assassin in Japan` participates in two senses. First, is the direct semantic sense as a definition of what a Ninja is. But there is a second, indirect sense: the word `Some` indicates that the solution lies as some substring of the phrase, namely `"assassi(n in Ja)pan"`. Thus, constructing the clue requires identifying a triangular relationship between $\{$`Ninja`, `(Some assassin in Japan)`$_1$, `(Some assassin in Japan)`$_2\}$ just like in an antanaclasis. This is true generally of any & Lit. clues as these clues must perform "double duty" in pointing to the answer.

B.4. Further evidence of our argument in §2.6

Below we provide two more pieces of evidence affirming the failure mechanism of next-token prediction outlined in §2.6.

Improved algorithmic creativity is not due to some form of capacity control. While §2.6 argues that multi-token prediction should help creativity by providing critical lookahead capabilities, it is also possible that it simply acts as a form of capacity control that prevents memorization. We rule this out in Fig 8: even as memorization computed on *unseen* seeds is controlled, the multi-token model perfectly reproduces the training data on *seen* seeds. We term this *seed-memorization*. An exact equivalence of this phenomenon was noticed in GANs in Nagarajan et al. (2018), where the generator can be trained on specific latent vectors to memorize the mapping on those, and yet produce fresh samples outside of those latent vectors.

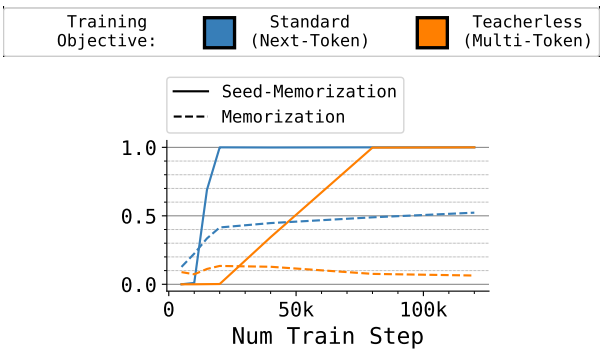

Figure 8. **Even if multi-token prediction reduces memorization (on unseen seeds), it has enough capacity to memorize training data on the seen seeds (denoted by seed-memorization).** Note that the best algorithmic creativity for `NTP` and `MTP` are achieved at step 10k and 40k, respectively, which are the checkpoints we used to report metrics in Fig 4.

Effect of token reordering. The implication of our argument in §2.6 is that next-token learning would benefit from reversing the token ordering of the `Sibling Discovery` task (i.e., parent appears before siblings). Indeed, we find this to be the case in Fig 12 and Fig 22. Interestingly, we find that the reverse-trained `NTP` model is still far from the original multi-token teacherless model. More surprisingly, a teacherless model trained on the reversed data, achieves even higher algorithmic creativity of all training methods here. Note that in all other datasets, no reordering of the tokens should make any change to the training.